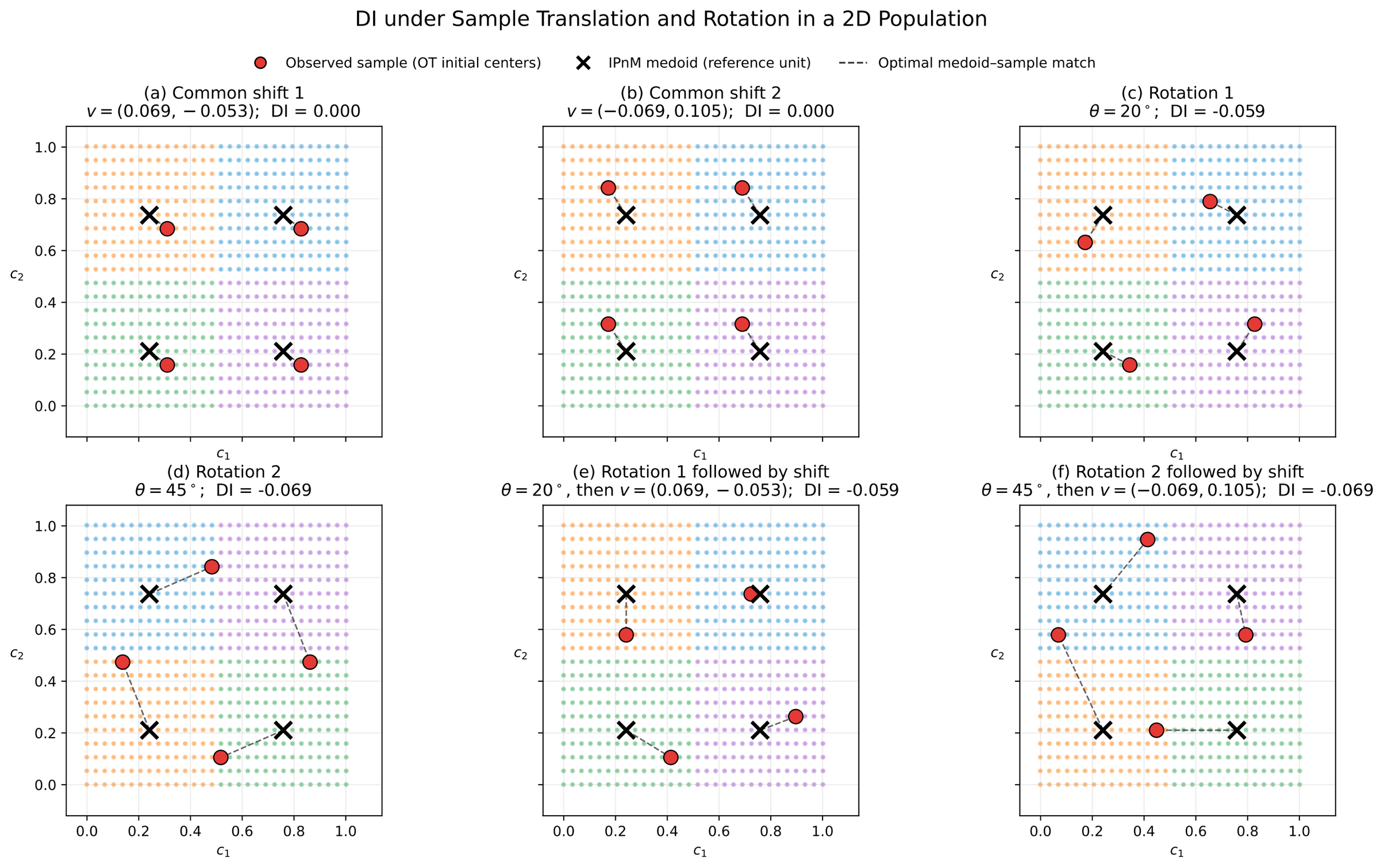

DI under Sample Translation and Rotation in a 2D Population
Observed sample (OT initial centers)
IPnM medoid (reference unit)
Optimal medoid–sample match
(a) Common shift 1
$v = (0.069, -0.053)$; DI = 0.000
(b) Common shift 2
$v = (-0.069, 0.105)$; DI = 0.000
(c) Rotation 1
$\theta = 20^\circ$; DI = -0.059
(d) Rotation 2
$\theta = 45^\circ$; DI = -0.069
(e) Rotation 1 followed by shift
$\theta = 20^\circ$, then $v = (0.069, -0.053)$; DI = -0.059
(f) Rotation 2 followed by shift
$\theta = 45^\circ$, then $v = (-0.069, 0.105)$; DI = -0.069
c_1
c_2
0.0
0.2
0.4
0.6
0.8
1.0